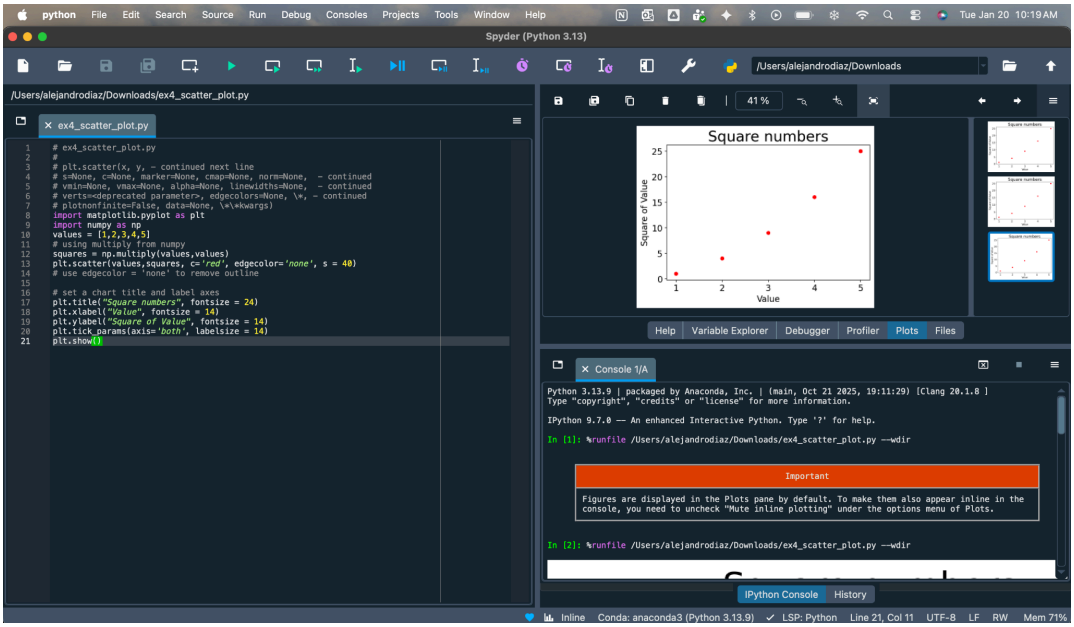
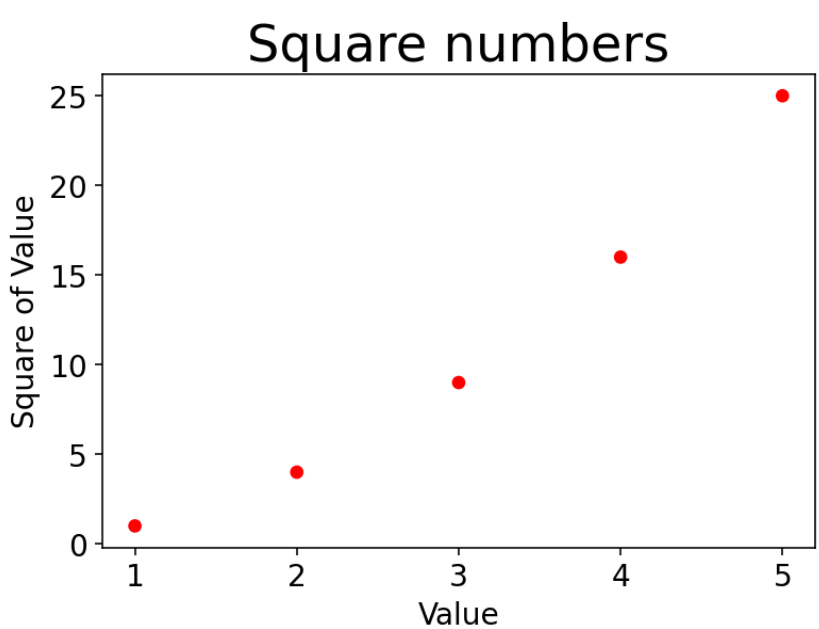
ME 144L Pre Lab: Introduction to DSCLab and Using Python

Python Installation

I have a working installation of Python, running on version 3.13.9.



Below is a provided example plot titled ‘Square numbers’, indicating that the Python installation and Spyder are functioning properly.



Phone Falling-Over Model

The Phone Falling-Over experiment can be described as a compound inverted pendulum, as it contains a distributed mass, a defined distance from the pendulum pivot to the center of mass, and a moment of inertia about the pivot. Using Newton's Second Law for fixed-axis rotation, the inverted pendulum equation can be described as shown:

$$\dot{h} = J\dot{\omega} = \sum M_o$$

Where the derivative of h is the rate of change of angular momentum, J is the moment of inertia of the pendulum object (phone). The derivative of ω represents the angular acceleration of the object, and the far right-hand side of the equation represents the sum of all external moments about the pivot point.

In the proposed system, it is essential to understand the critical assumptions before considering the involved moments. In this simulation, the underlying assumptions are that the contact point between the phone and the floor acts as a fixed pivot and energy losses from air resistance are negligible. For this pendulum, it will assume that mass is uniformly distributed across the length of the phone, and thus the center of gravity is at half the total length of the pendulum.

Solving the system in terms of the state variables 'ω' and 'θ' allows for the quantification of total energy in the system, where 'ω' is used to quantify kinetic energy and 'θ' represents the potential energy based on the phone's position relative to the floor and gravitational field.

For this model, the primary external torque is the moment induced from gravity, applied at the center of mass halfway down the length of the phone. The moment can be described as a function of angular position, θ. The derivative of θ is equal to the angular velocity of the object; the moment and angular acceleration equations can be combined. This creates the following set of equations:

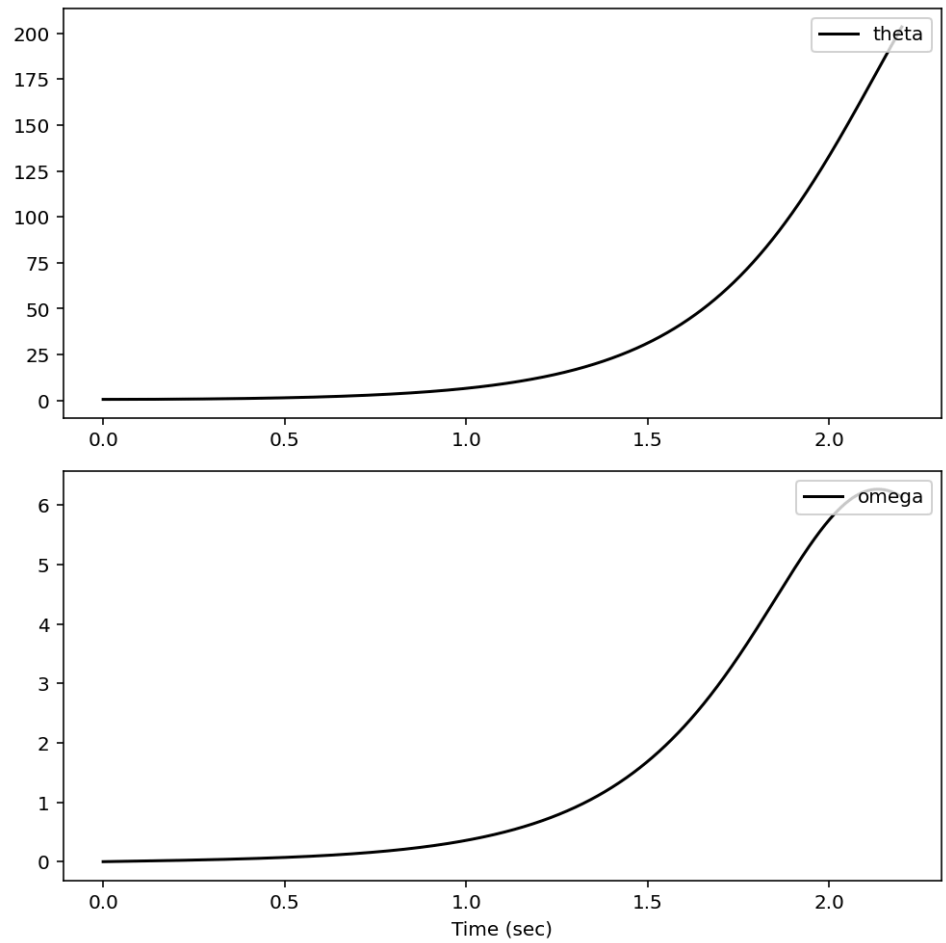
$$J\dot{\omega} = \sum M_o = mg\left(\frac{L}{2}\right)\sin(\theta)$$

$$\dot{\theta} = \omega$$

To simulate the system, the second-order rotational equation is converted into two first-order Ordinary Differential Equations (ODEs). Using the equations from above and rearranging variables to solve for angular acceleration creates the following state space in vectorized form:

$$\begin{bmatrix} \dot{\omega} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} mg\left(\frac{L}{2J}\right)\sin\theta \\ \omega \end{bmatrix}$$

To initiate the simulation and match experimental release from a vertical position:

- $\omega(0) = 0$: The phone is released from rest.
- $\theta(0) \approx 0$: A very small, non-zero initial angle is required. If it were perfectly zero, the model would remain balanced and never fall.

Simulation Results